

Chapter-1

- 1. **Define software.** Software is a set of instructions, programs, and data that directs a computer's actions; it is a program or set of programs designed to provide specific desired functionality .
- 2. **Why are software myths harmful?** They are harmful because they propagate misconceptions that can lead to unrealistic expectations, poor decision-making, and inefficient workflows for stakeholders, including managers and developers .
- 3. **List any two umbrella activities of software process.** Any two of the following: **Software Project Tracking and Control, Risk Management, Software Quality Assurance (SQA),** or **Software Configuration Management (SCM)** .
- 4. **State the evolving role of software in modern systems.** Software serves a **dual role:** as a **product** (transforming information and delivering computing potential) and as a **vehicle** for delivering that product (controlling other software or helping build new software tools) , .
- 5. **What is process adaptation?** Process adaptation is the ability of a process to modify its behavior or structure to better suit new conditions, requirements, or changes in its environment .

Chapter-2

- 1. **What is reverse engineering?** Reverse engineering is the process of extracting design information from source code to understand threat mechanisms (in cybersecurity) or to find cost-effective alternatives for components , .
- 2. **List the advantages of Concurrent Model.** It is applicable to all software development types, easy to understand, provides immediate feedback from testing, and offers an accurate picture of a project's current state .
- 3. **Define generic process model and list the phases of generic process model.** The generic process model is an abstraction of the software development process . Its five phases are **Communication, Planning, Modelling, Construction,** and **Deployment** .
- 4. **What is unified process?** The Unified Process (UP) is an iterative, architecture-centric, and use-case-driven methodology that emphasizes collaboration, risk management, and incremental delivery , .

Chapter-3

- 1. **List the advantages of waterfall model.** It is simple to use and manage due to its rigid structure, has clearly defined stages and milestones, and works well for small projects with stable, well-understood requirements , .
- 2. **Define prescriptive process model.** These are traditional models that follow a structured, sequential approach to development to ensure predictability, discipline, and organization .
- 3. **What is meant by human factors in agile development?** Human factors refer to team dynamics such as **collaboration, mutual trust, self-organization,** and a **common focus** on delivering a working software increment , , .
- 4. **Explain Dynamic System Development Model with neat diagram.** DSDM is an iterative Agile framework providing guidance throughout the project life cycle to deliver products on time and within budget .
Note: The source text describes the model but does not provide a visual diagram for reproduction here.
- 5. **Name any two key XP activities.** Any two of: **Planning, Designing, Coding, Testing,** or **Listening** , .
- 6. **Define agility.** Agility refers to being quick and adaptable; in software, it is an approach based on iterative development that breaks projects into small sections to handle changing requirements .
- 7. **What are the advantages of Scrum?** Scrum is fast-moving, cost-efficient, increases quality through constant feedback, and achieves high customer satisfaction by dividing large products into smaller sub-products .
- 8. **List any two agile principles.** Any two of: **Early and continuous delivery of valuable software, welcoming changing requirements,** or **frequent delivery of working software** .

Chapter-4

- 1. **State the purpose of use case diagram.** Use case diagrams are used to drive development and represent the project architecture, helping developers and customers gain insight into the end product , .
- 2. **Write the symbols of activity diagram.** *Note: The provided sources mention UML modeling generally but do not explicitly list the specific visual symbols for activity diagrams* .
- 3. **What is elicitation?** Elicitation is the process of identifying requirements by interviewing stakeholders, reviewing documents, and identifying missing or ambiguous information .
- 4. **What is negotiation?** Negotiation involves communicating with end users to analyze their views and reconcile different stakeholder requirements to form a clear specification .
- 5. **What is requirement validation?** Requirement validation is the process of reviewing requirements to ensure they are complete, consistent, and correctly define the user's needs before development proceeds , .

Chapter-5

- 1. **Write any two testing strategies?** Any two of: **Black box testing, White box testing, Unit testing,** or **Integration testing** , .
- 2. **What is black box testing?** It is a method where the tester evaluates the external behavior and functionality of the software without knowing its internal code structure or design , .
- 3. **What is white box testing.** It is a method where the tester has access to the internal code and logic, using this knowledge to verify correctness at the code level , .
- 4. **Define system testing.** System testing is the evaluation of the complete, integrated software system to ensure it meets specified functional and technical requirements , .
- 5. **Define unit testing.** Unit testing involves testing individual components (like functions or objects) in isolation to ensure they function correctly before being integrated into larger units , .
- 6. **List the phases of STLC.** The phases are **Requirement Analysis, Test Planning, Test Case Development, Test Environment Setup, Test Execution,** and **Test Closure**